

Madhya Pradesh – Grade 10

2019 – Mathematics

Choose the correct option and write it in your answer book

1. The H.C.F of 96 and 404 is:

- (a) 120
- (b) 4
- (c) 10
- (d) 3

Answer: (b)

Solution:

$$\begin{array}{r} 96 \overline{) 404} \\ \underline{384} \\ 20 \\ \underline{80} \\ 16 \\ \underline{16} \\ 4 \\ \underline{4} \\ 0 \end{array}$$

2. If α and β are the zeroes of the quadratic polynomial $ax^2 + bx + c$, then the value of $\alpha.\beta$ is:

- (a) $\frac{c}{a}$
- (b) $\frac{a}{c}$
- (c) $\frac{-c}{a}$
- (d) $\frac{-a}{c}$

Answer: (a)

3. The zeroes of the polynomial $x^2 - 3$ will be

- (a) $\pm\sqrt{3}$
- (b) ± 3
- (c) 3
- (d) 9

Answer: (a)

Solution:

$$x^2 - 3 = 0$$

$$x = \pm\sqrt{3}$$

4. When $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ then the system of equation $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

- (a) Has two solutions
- (b) Has no solution
- (c) Has infinitely many solutions
- (d) Has unique solution

Answer: (b)

5. Lines $x - 2y = 0$ and $3x + 4y - 20 = 0$ are:

- (a) Intersect
- (b) Coincide
- (c) Parallel
- (d) None

Answer: (a)

Solution:

$$a_1 = 1 \quad a_2 = 3$$

$$b_1 = -2 \quad b_2 = 4$$

$$c_1 = 0 \quad c_2 = -20$$

$$\frac{a_1}{a_2} = \frac{1}{3} \quad \frac{b_1}{b_2} = \frac{-2}{4} \quad \frac{c_1}{c_2} = \frac{0}{-20}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \text{ Hence Intersecting}$$

Fill in the blanks.

1. A quadratic equation $ax^2 + bx + c = 0$ has no real root if

Answer: $\Delta < 0$

2. The discriminant of the equation $3x^2 - 2x + \frac{1}{3} = 0$ is

Answer: $\Delta = 0$

$$a = 3 \quad b = -2 \quad c = \frac{1}{3}$$

$$b^2 - 4ac$$

$$4 - 4 \times 3 \times \frac{1}{3} = 0$$

3. In A.P $\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}, \dots$ the common difference d is

Answer: -1

4. The sum of the probabilities of all the elementary events of an experiment is.

Answer: 1

5. Formula of area of the sector of angle θ is

Answer: $\frac{\theta}{360} \times \pi r^2$

Write true/false in the following:

1. The perpendicular drawn from the centre of a circle to a chord bisect the chord.

Answer: True

2. All squares are similar.

Answer: True

3. Area of right triangle = $\frac{1}{2} \times \text{base} \times \text{altitude}$.

Answer: True

4. A line intersecting a circle in two points is called a secant.

Answer: True

5. The angle of elevation of an object viewed is the angle formed by the line of sight with the horizontal, when we lower our head to look at the object.

Answer: False

Write the answer in one word/sentence.

1. What will be the Arithmetic mean of 1, 2, 3, 4, 5?

Answer: 3

Solution:

$$\frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

2. Write the formula of the median.

Answer: Median = $l + \frac{\left(\frac{n}{2} - Cf\right)}{f} \times h$

3. Find the value of probability of an EVENT E + probability of the EVENT "NOT E"

Answer: 1

4. Write the formula of volume of a frustum a cone.

Answer: $\frac{1}{3} \pi h (r_1^2 + r_2^2 + r_1 r_2)$

5. How many parallel tangents of a circle?

Answer: 2

Match the correct column

Column 'A'	Column 'B'
(1) $1 + \cot^2 \theta$	(a) $\sin \theta$
(2) $\sec \theta$	(b) 0
(3) $\sin^2 \theta + \cos^2 \theta$	(c) $\sqrt{3}$
(4) $\tan 60^\circ$	(d) 1
(5) $\cos(90 - \theta)$	(e) $\operatorname{cosec}^2 \theta$
	(f) $\frac{1}{\cos \theta}$
	(g) $\frac{1}{\sqrt{3}}$

Answer:

- 1 – e
- 2 – f
- 3 – d
- 4 – c
- 5 – a

Question: Find the LCM and HCF of 6 and 20 by the prime factorisation method.

OR

Find the H.C.F of 6, 72 and 120 using the prime factorisation method.

Solution:

(a) $6 = 2 \times 3$

$20 = 2 \times 2 \times 5$

HCF = 2

LCM = $2 \times 2 \times 3 \times 5 = 60$

(b) $6 = 2 \times 3$

$72 = 2 \times 2 \times 2 \times 3 \times 3$

$120 = 2 \times 2 \times 2 \times 3 \times 5$

HCF = 2×3

Question: Find a quadratic polynomial, the sum and product of whose zeroes are -3 and 2 respectively.

OR

Divide $2x^2 + 3x + 1$ by $x + 2$.

Solution:

$$(a) \alpha + \beta = -3 = \frac{-b}{a}$$

$$\alpha\beta = 2 = \frac{c}{a}$$

$$\text{Equation} = x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$= x^2 + 3x + 2 = 0$$

(b)

$$x + 2 \quad 2x^2 + 3x + 1 \quad (2x - 1)$$

$$\begin{array}{r} 2x^2 + 4x \\ - \quad - \\ \hline -x + 1 \\ -x - 2 \\ + \quad + \\ \hline 3 \end{array}$$

Question: Find the distance between points (2, 3) and (4, 1)

OR

Find the area of a triangle whose vertices (1, -1), (-4, 6) (-3, -5)

Solution:

$$(a) \text{ Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$(2, 3) = x_1 = 2$$

$$(4, 1) = x_2 = 4$$

$$y_1 = 3$$

$$y_2 = 1$$

$$PQ = \sqrt{(4-2)^2 + (3-1)^2}$$

$$= \sqrt{4+4} = 2\sqrt{2}$$

(b)

$$\text{Area} = \frac{1}{2} \{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\}$$

$$= \frac{1}{2} [1(6+5) + (-4)(-5+1) + (-3)(-1-6)]$$

$$= \frac{1}{2} [11+16+21] = 24$$

Question: Two players Sangeeta and Reshma, play a tennis match. It is known that the probability of Sangeeta winning the match is 0.62. What is the probability of Reshma winning the match.

OR

A box contains 3 blue, 2 white and 4 red marbles. If a marble is drawn at random from the box, what is the probability that will be a (i) white? (ii) blue? (iii) red ?

Solution:

(a) Sangeeta probability of winning $P(s) = 0.62$

Sangeeta Probability of losing = $1 - 0.62 = 0.38$

Sangeeta probability of losing = Rashmi probability of winning
= 0.38

(b)

Total marbles = $3 + 2 + 4 = 9$

Probability of white = $\frac{2}{9}$

Probability of Blue = $\frac{3}{9} = \frac{1}{3}$

Probability of Red = $\frac{4}{9}$

Question: If $P(E) = 0.05$, what is the probability of ("not E") i.e. $P(\bar{E})$?

OR

One card is drawn from a well – shuffled deck of 52 cards. Calculate the probability, that the card will

(i) be an ace

(ii) not be an ace.

Solution:

(a)

$$\begin{aligned} P(\bar{E}) &= 1 - P(E) \\ &= 1 - 0.05 \\ &= 0.95 \end{aligned}$$

(b)

Total cards = 52

Aces = 4

Probability of Ace = $\frac{4}{52} = \frac{1}{13}$

Probability of Not Ace = $1 - \frac{1}{13} = \frac{12}{13}$

Question: Prove that : $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$

OR

Evaluate the following: $\sin 60^\circ \cdot \cos 30^\circ + \sin 30^\circ \cdot \cos 60^\circ$

Solution:

(a) L.H.S = $\sqrt{\frac{1+\sin A}{1-\sin A}} \times \frac{1+\sin A}{1+\sin A}$

$$= \sqrt{\frac{(1+\sin A)^2}{1-\sin^2 A}}$$

$$= \frac{1+\sin A}{\cos A} = \sec A + \tan A = \text{R.H.S}$$

(b)

$$\sin 60 \times \cos 30 + \sin 30 \times \cos 60$$

$$= \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{3}{4} + \frac{1}{4}$$

$$= 1$$

Question: Find the Value of K, if the points A(2, 3), B(4, k) and C(6, -3) are collinear.

OR

Find the ratio which the y- axis divided the line segment joining the points (5, -6) and (-1, -4) also find the point of intersection.

Solution:

(a) For collinearity Area of triangle is zero

$$= \frac{1}{2} [2(k+3) + 4(-3-3) + 6(3-k)] = 0$$

$$= \frac{1}{2} (-4k) = 0$$

$$k = 0$$

(b)

let the ratio be $k : 1$

$$\text{Point of intersection} = \left(\frac{-k+5}{k+1}, \frac{-4k-6}{k+1} \right)$$

$\therefore$ The line is intersected by y axis so, $x = 0$

$$\frac{-k+5}{k+1} = 0$$

$$k = 5$$

Hence, ratio is $\underline{k} : 1$ or $5 : 1$

$$\text{Point is } \left(0, \frac{-13}{3} \right)$$

Question: The length of tangent drawn from an external point to a circle are equal.

OR

Prove that the tangent drawn at the ends of a diameter of a circle are parallel.

Solution:

(a) In $\triangle ABO$ and $\triangle ACO$

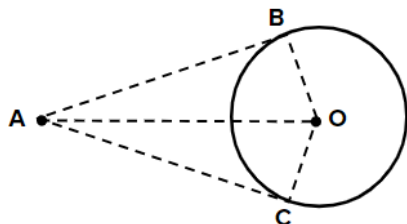
AO = AO (common)

OB = OC (Radii)

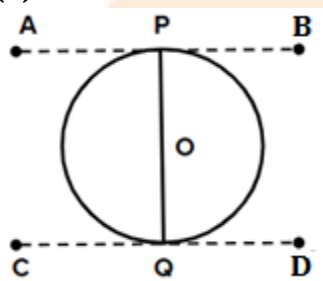
$$\angle ABO = \angle ACO = 90^\circ$$

Hence, $\triangle ABO \cong \triangle ACO$

$AB = AC$ by c.p.c.t



(b)



$$\therefore OP \perp AB$$

$$\angle OPB = 90^\circ$$

$$\therefore OQ \perp CD$$

$$\angle OQC = 90^\circ$$

$$\angle OPB + \angle OQC = 90 + 90 = 180^\circ$$

Hence, alternate interior angle are equal $AB \parallel CD$

Question: Find the area of the sector of a circle with radius 4 cm and angle 30° . Also find the area of the corresponding major sector. (Use $\pi = 3.14$)

OR

Find the area of sector of a circle with radius 6 cm whose angle of sector is 60°

Solution:

$$(a) \text{ Area of sector} = \frac{\theta}{360} \times \pi r^2$$

$$= \frac{\theta}{360} \times 3.14 \times 4 \times 4 \text{ cm}^2$$

$$= 4.19 \text{ cm}^2$$

$$\text{Total area} = \pi r^2$$

$$= 3.14 \times 4 \times 4$$

$$= 50.24 \text{ cm}^2$$

$$\text{Area of major segment} = 50.24 - 4.19$$

$$= 46.1 \text{ cm}^2$$

(b) $r = 6 \text{ cm}$ $\theta = 60^\circ$

$$\begin{aligned}\text{Area} &= \frac{\theta}{360} \times \pi r^2 \\ &= \frac{60}{360} \times 3.14 \times 6 \times 6 \\ &= \frac{132}{7} \text{ cm}^2\end{aligned}$$

Question: Prove that $5 - \sqrt{3}$ is irrational number.

OR

Show that any positive odd integer is of the form $4q + 1$ or $4q + 3$ where q is an integer.

Solution:

(a) Let $5 - \sqrt{3}$ be a rational so, $5 - \sqrt{3} = \frac{a}{b}$

$$5 - \frac{a}{b} = \sqrt{3}$$

Since, $\frac{a}{b}$ is a rational number and any operation with rational gives rational only.

So, $5 - \frac{a}{b}$ = rational also

$\sqrt{3}$ = must be rational also

But since, $\sqrt{3}$ is irrational

Hence, case fails so, $5 - \sqrt{3}$ is irrational

(b) Let a be any integer where $b = 4$

By Euclid algorithm $a = 4q + r$

Such that $q > 0 \dots r = 0, 1, 2, 3$ because $0 \leq r < 4$

$a = 4q, 4q + 1, 4q + 2, 4q + 3$

if $a = 4q, 4q + 2$ so, it's is an even

so, by Euclid,

$4q + 1$ and $4q + 3$ is a positive integer

Question: Find the zeros of the quadratic polynomial $x^2 + 7x + 10$ and verify the relationship between the zeros and the coefficients.

OR

Divide $3x^2 + x^2 + 2x + 5$ by $1 + 2x + x^2$.

Solution:

(a)

$$x^2 + 7x + 10 = 0$$

$$x^2 + 5x + 2x + 10 = 0$$

$$(x+5)(x+2) = 0$$

$$x = -5 \text{ or } x = -2$$

$$\therefore \alpha + \beta = \frac{-b}{a} = \frac{-7}{1} = -7$$

Roots = $-5 + -2 = -7$ (Hence, sum of roots are equal)

$$\alpha \times \beta = \frac{c}{a} = 10 = -5 \times -2$$

(Hence verified)

(b)

$$x^2 + 2x + 1 \quad 3x^3 + x^2 + 2x + 5 \quad (3x - 5)$$

$$\begin{array}{r} 3x^3 + 6x^2 + 3x \\ - \quad - \quad - \\ \hline -5x^2 - x + 5 \\ -5x^2 - 10x - 5 \\ + \quad + \quad + \\ \hline 9x + 10 \end{array}$$

Question: If the sum of the first 14 terms of an A.P. is 1050 and its first term is 10. Find the 20th term.

OR

Find the 31st term of an A.P. whose 11th term is 38 and 16th term is 73.

Solution:

(a) $S_{14} = 1050$, $n = 14$, $a = 10$ are given

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$1050 = \frac{14}{2} [2 \times 10 + (14-1) \times d]$$

$$910 = 91d$$

$$d = 10$$

$$\text{Now, } a_{20} = a_0 + (n-1)d$$

$$= 10 + 19 \times 10$$

$$= 200$$

(b)

According to equation

$$a_{11} = 38 \text{ and } a_{16} = 73$$

$$a_{11} = a + (n-1)d$$

$$38 = a + 10d \dots (1)$$

$$a_{16} = a + 15d$$

$$73 = a + 15d \dots (2)$$

Solving (1) and (2)

$$5d = 35$$

$d = 7$ put in equation (1)

$$a + 10 \times 7 = 38$$

$$a = -32$$

$$\text{Now, } a_{31} = a + 30d$$

$$= -32 + 30 \times 7$$

$$= 178$$

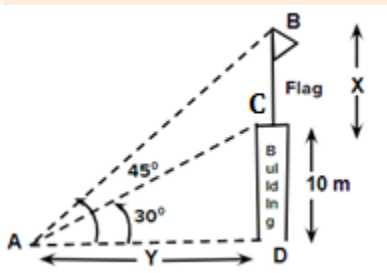
Question: From a point A on the ground the angle of elevation of the top of a 10-meter-tall building is 30° . A flag is hosted at the top of the building and the angle of elevation of the top of the flagstaff from A is 45° . Find the length of the flagstaff and the distance of the building from the point A. (you may take $\sqrt{3} = 1.732$)

OR

The angle of depression of the top and the bottom of an 8 m. tall building from the top of a multi – storeyed building are 30° and 45° respectively. Find the height of the multi-storeyed building and distance between the two buildings.

Solution:

(a)



Let length of flag be x distance point is y

In $\triangle ACD$

$$\tan 30 = \frac{10}{y}$$

$$y = 10\sqrt{3}$$

$$= 10 \times 1.732 = 17.32 \text{ m}$$

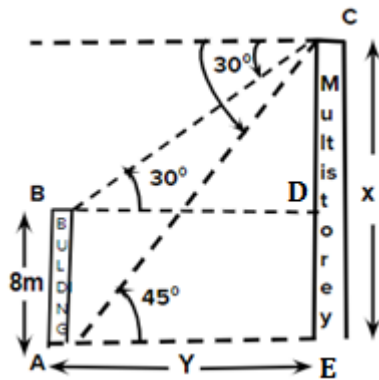
In $\triangle ABD$

$$\tan 45 = \frac{10 + x}{17.32}$$

$$17.32 = 10 + x$$

$$x = 7.32 \text{ m}$$

(b)



Let height of multi-storeyed building be x distance between building = y

In $\triangle BCD$

$$\tan 30^\circ = \frac{CD}{BD}$$

$$\frac{1}{\sqrt{3}} = \frac{x-8}{y}$$

$$y = \sqrt{3}x - 8\sqrt{3} \dots \dots (1)$$

In $\triangle ACE$

$$\tan 45^\circ = \frac{CE}{AE} = \frac{x}{y}$$

$$(x = y) \dots \dots \dots (2)$$

Put equation (2) in equation (1)

$$x = \sqrt{3}x - 8\sqrt{3}$$

$$8\sqrt{3} = \sqrt{3}x - x$$

$$8 \times 1.732 = 0.732x$$

$$x = 18.92 \text{ m}$$

$$y = 18.92 \text{ m}$$

Question: Use Elimination method to find all possible solutions of the following pair of linear equation:

$$2x + 3y = 8$$

$$4x + 6y = 7$$

OR

The cost of 5 oranges and 3 apples is Rs. 35 and the cost of 2 oranges and 4 apples is Rs. 28.

Let us find the cost of an orange and an apple.

Solution:

(a)

$$2x + 3y = 8 \dots (1)$$

$$4x + 6y = 7 \dots (2)$$

Multiply equation (1) with 2

$$4x + 6y = 16 \dots \dots \dots (3)$$

$$4x + 6y = 7 \dots\dots\dots(4)$$

Since is not possible to solve (3) and (4)

Let's check the condition

$$\frac{a_1}{a_2} = \frac{4}{4} \quad \frac{b_1}{b_2} = \frac{6}{6} \quad \frac{c_1}{c_2} = \frac{16}{7}$$

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \text{ (parallel lines)}$$

Hence, no solution.

(b)

Let cost of orange be x and apple be y

$$5x + 3y = 35 \times 4$$

$$2x + 4y = 28 \times 3$$

$$20x + 12y = 140$$

$$6x + 12y = 84$$

$$\begin{array}{r} - \quad - \quad - \\ \hline \end{array}$$

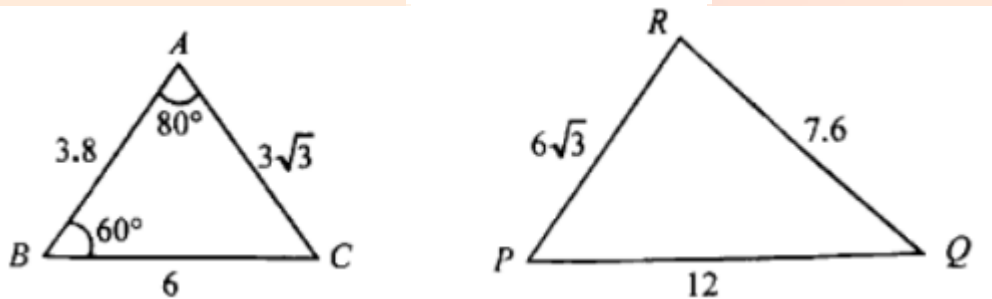
$$14x = 56$$

$$x = 4 \text{ _____ orange cost}$$

$$y = \frac{35 - 5x}{3}$$

$$= \frac{35 - 5x}{3} = \frac{35 - 20}{3} = \frac{15}{3} = 5 \text{ _____ Apple cost}$$

Question: Observe in fig. find $\angle P$



OR

ABC is an Isosceles triangle right angled at C. Prove that $AB^2 = 2AC^2$.

Solution:

(a)

$$\frac{AB}{RQ} = \frac{3.8}{7.6} = \frac{1}{2} \quad \frac{BC}{QP} = \frac{6}{12} = \frac{1}{2} \quad \frac{CA}{PR} = \frac{3\sqrt{3}}{6\sqrt{3}} = \frac{1}{2}$$

$$\text{It means } \frac{AB}{RQ} = \frac{BC}{QP} = \frac{CA}{PR}$$

$\triangle ABC \sim \triangle RQP$ BySSS

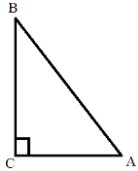
Hence, $\angle C = \angle P$

$$\angle C = 180 - \angle A - \angle B$$

$$= 40^\circ$$

$$\angle P = 40^\circ$$

(b)



$$BC = AC$$

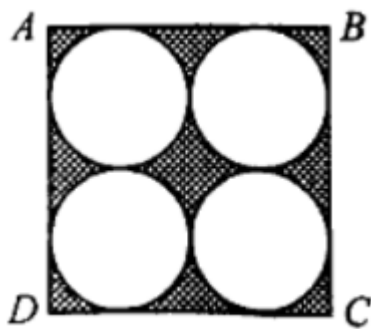
$$AB^2 = BC^2 + AC^2 \quad (\text{By Pythagoras})$$

$$AB^2 = AC^2 + AC^2$$

$$= 2AC^2$$

Hence proved

Question: Find the area of the shaded region in given figure where ABCD is a square of side 14 cm.



OR

In a circle of radius 21 cm an arc subtends an angle of 60° at the centre find; (i) the length of the arc (ii) area of the sector.

Solution:

(a)

$$\text{Area of } ABCD = 14 \times 14 = 196 \text{ cm}^2$$

$$\text{Radius} = \frac{1}{2} \text{ of } \frac{1}{2} \text{ of square}$$

$$\therefore \text{diameter} = \frac{1}{2} \text{ of square side}$$

$$= \frac{1}{2} \times \frac{1}{2} \times 14 = \frac{7}{2} \text{ cm}$$

$$\text{Area of one circle} = \pi \times \frac{7}{2} \times \frac{7}{2}$$

$$= \frac{49}{4} \pi \text{ cm}^2 = \frac{77}{2} \text{ cm}^2$$

$$\text{Area of 4 circle} = \frac{77}{2} \times 4 = 154 \text{ cm}^2$$

$$\text{Area of shaded region} = 196 - 154 = 42 \text{ cm}^2$$

(b) (i) Length of arc

$$= \frac{\theta}{360} \times 2\pi r$$

$$= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22 \text{ cm}$$

$$\text{(ii) Area of sector} = \frac{\theta}{360} \times \pi r^2$$

$$= \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = 231 \text{ cm}^2$$

Question: Find the roots of the following equation: $x + \frac{1}{x} = 3, x \neq 0$

OR

Find two consecutive odd positive integers, sum of whose squares is 290.

Solution:

(a)

$$x + \frac{1}{x} = 3$$

$$x^2 + 1 = 3x$$

$$x^2 - 3x + 1 = 0$$

$$a = 1 \quad b = -3 \quad c = 1$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \times 1 \times 1}}{2 \times 1}$$

$$\text{using } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{3 \pm \sqrt{9 - 4}}{2} = \frac{3 + \sqrt{5}}{2}, \frac{3 - \sqrt{5}}{2}$$

(b) let the odd integers be x and $x + 2$

$$x^2 + (x+2)^2 = 290$$

$$x^2 + x^2 + 4 + 4x = 290$$

$$2x^2 + 4x - 286 = 0$$

$$x^2 + 2x - 143 = 0$$

$$x^2 + 2x - 143 = 0$$

$$x = \frac{-2 \pm \sqrt{4 + 4 \times 1 \times 143}}{2}$$

$$= \frac{-2 \pm \sqrt{576}}{2} = \frac{-2 \pm 24}{2}$$

$$= \frac{-2 + 24}{2}, \frac{-2 - 24}{2}$$

$$= 11, -13$$

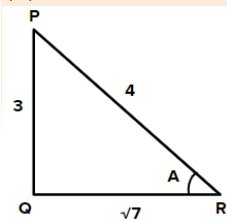
Question: If $\sin A = \frac{3}{4}$, calculate $\cos A$ and $\tan A$

OR

Evaluate the following: $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$

Solution:

(a)



Let in $\triangle PQR$, A angle is there by Pythagoras

$$QR = \sqrt{4^2 - 3^2}$$

$$= \sqrt{7}$$

$$\cos A = \frac{\sqrt{7}}{4} \quad \tan A = \frac{3}{\sqrt{7}}$$

(b)

$$2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$$

$$= 2 \times (1)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2$$

$$= 2 + \frac{3}{4} - \frac{3}{4} = 2$$

Question: Construct a triangle similar to a given triangle ABC with side equal to $\frac{5}{3}$ of the corresponding sides of the triangle ABC.

OR

Draw a circle with the help of bangle. Take a point outside the circle. Construct the pair of tangents from this point to the circle.

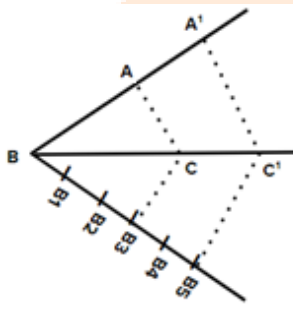
Solution:

(a)

Given $\triangle ABC$, we are required to construct a triangle whose sides are of the corresponding sides of $\triangle ABC$.

Steps are:

- Draw any ray BX making an acute angle with BC on side opposite to vertex A.
 - Locate the points B_1, B_2, B_3, B_4, B_5 on BX such that $BB_1 = B_1B_2 = B_2B_3 = B_3B_4 = B_4B_5$
 - Join B_3 to C as 3 being smaller and through B_5 draw a line parallel to B_3C , intersecting the extended line segment BC at C'
 - Draw a line through C' parallel to CA intersecting the segment BA at A'
- $A'B'C'$ is required triangle

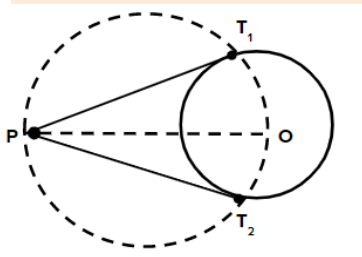


(b)

Draw a circle

Place a point outside the circle named P.

- Join P and centre O. Draw its perpendicular bisector
- Draw the circle and Join tangent.



Question: The radii of the ends of a frustum of a cone 45 cm high are 28 cm and 7 cm respectively, find the volume.

OR

A hemispherical tank full of water is emptied by a pipe at the rate of $3\frac{4}{7}$ litre per second. How much time will it take to empty half the tank, if it is 3 m in diameter? (Take $\pi = \frac{22}{7}$)

Solution:

(a) Volume of frustum = $\frac{1}{3}\pi h(r_1^2 + r_2^2 + r_1r_2)$

$$r_1 = 28\text{cm} \quad r_2 = 7\text{cm} \quad h = 45\text{cm}$$

$$V = \frac{1}{3} \times \frac{22}{7} \times 45 (28^2 + 7^2 + 28 \times 7)$$

$$= 48510\text{cm}^3$$

(b) Radius of tank = $\frac{3}{2}\text{m}$

$$\text{Volume of hemisphere tank} = \frac{2}{3} \pi r^3$$

$$= \frac{2}{3} \times \frac{22}{7} \times \frac{3}{2} \times \frac{3}{2} \times \frac{3}{2}$$

$$= \frac{99}{14} \text{m}^3$$

$$= \frac{99}{14} \times 1000 = \frac{99000}{14} \text{litre}$$

Water empties at $\frac{25}{7}\text{L}$ per second

So, $\frac{99000}{14}\text{L}$ empties in $\frac{99000}{14} \times \frac{7}{25}$

$$= 1980 \text{ second}$$

$$= 33 \text{ minutes.}$$

Question: A survey conducted on 20 households in locality by a group of students resulted in the following frequency table for the number of family member in a household:

Family size	1-3	3 - 5	5 - 7	7 - 9	9 - 11
Number of families	7	8	2	2	1

Find the Mode of this data

OR

In given data,

Class Interval	Number of students
10 - 25	2
25 - 40	3
40 - 55	7
55 - 70	6
70 - 85	6
85 - 100	6

Find the Mean.

Solution:

Family Size	No. of families
1 - 3	7
3 - 5	8
5 - 7	2
7 - 9	2
9 - 11	1

Modal class = 3 - 5 as it has the highest frequency of 8

$$\text{Now mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

l = lower limit of modal class

f_1 = frequency of modal class

f_0 = frequency of proceed class

f_2 = frequency of succeeding class

$$l = 3$$

$$f_1 = 8$$

$$f_0 = 7$$

$$f_2 = 2$$

$$\text{Mode} = 3 + \left(\frac{8 - 7}{2 \times 8 - 7 - 2} \right) \times 2$$

$$= 3 + \left(\frac{1}{7} \right) \times 2$$

$$= 3.286$$

(b)

Class Interval	frequency	Marks x_i	$f_i x_i$
10 - 25	2	17.5	35
25 - 40	3	32.5	97.5
40 - 55	7	47.5	332.5
55 - 70	6	62.5	375
70 - 85	6	77.5	465
85 - 100	6	92.5	555
	$\Sigma f_i = 30$		$\Sigma f_i x_i = 1860$

$$\text{So, mean} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{1860}{30} = 62$$